THE NAIVE COTANGENT COMPLEX COMPUTED FROM A FACTORIZATION THROUGH A FORMALLY SMOOTH MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3979-3989 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3991-4015, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The naive cotangent complex, conormal sheaves of immersions, and formally smooth morphisms; the comparison is retained.

Source reference. Original lemma lemma-NL-formally-smooth: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3824–3830. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-NL-immersion: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3903–3907. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-smooth-formally-smooth: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3190–3197. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E1Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4015–4015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FUY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4015–4015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes which factors as $f = g \circ i$ with i an immersion and $g : P \to Y$ formally smooth (for example smooth). Then there is a canonical isomorphism*

$$NL_{X/Y} \cong \left(\mathcal{C}_{X/P} \to i^*\Omega_{P/Y}\right)$$

in $D(\mathcal{O}_X)$ where the conormal sheaf $\mathcal{C}_{X/P}$ is placed in degree -1.

Proof. (For the parenthetical statement see Lemma lemma-smooth-formally-smooth.) By Lemmas lemma-NL-immersion and lemma-NL-formally-smooth we have $NL_{X/P} = \mathcal{C}_{X/P}[1]$ and $NL_{P/Y} = \Omega_{P/Y}$ with $\Omega_{P/Y}$ locally projective. This implies that $i^* NL_{P/Y} \to i^*\Omega_{P/Y}$ is a quasi-isomorphism too (small detail omitted; the reason is that $i^* NL_{P/Y}$ is the same thing as $\tau_{\geq -1} Li^* NL_{P/Y}$, see More on Algebra, Lemma 0FUY). Thus the canonical map

$$i^* NL_{P/Y} \to \text{Cone}(NL_{X/Y} \to NL_{X/P})[-1]$$

of Modules, Lemma 0E1Z is an isomorphism in $D(\mathcal{O}_X)$ because the cohomology group $H^{-1}(i^* NL_{P/Y})$ is zero by what we said above. In other words, we have a distinguished triangle

$$i^* NL_{P/Y} \to NL_{X/Y} \to NL_{X/P} \to i^* NL_{P/Y}[1]$$

Clearly, this means that $NL_{X/Y}$ is the cone on the map $NL_{X/P}[-1] \to i^* NL_{P/Y}$ which is equivalent to the statement of the lemma by our computation of the cohomology sheaves of these objects in the derived category given above. $\square$